# Tirgul 2

## Exercise 2.6

**Theorem:** Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms ($a_n > 0$). If the limit superior of the ratio of successive terms is less than 1, specifically:

$$\overline{\lim}_{n\to\infty} \frac{a_{n+1}}{a_n} = L < 1$$

then the series $\sum_{n=1}^{\infty} a_n$ converges.

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### Definition of the Limit Superior ($\limsup$)

The limit superior (denoted by $\overline{\lim}$ or $\limsup$) of a sequence $\{x_n\}$ represents its eventual upper bound. Formally:

$$\overline{\lim}_{n\to\infty} x_n = \lim_{n\to\infty} \left( \sup_{k\geq n} x_k \right)$$

A critical property of the $\limsup$ is that for any $\epsilon > 0$, there are only a finite number of terms in the sequence $x_n$ that satisfy $x_n \geq L + \epsilon$. Consequently, there exists some index $N$ such that for all $n \geq N$, the terms satisfy $x_n < L + \epsilon$.

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1. **Applying the Definition:** By the definition of the limit superior, the inequality $\left|\frac{a_{n+1}}{a_n} - L\right| < \epsilon$ fails to hold only for a finite number of $n$. This implies that there exists a natural number $N$ such that for all $n \geq N$:

$$\frac{a_{n+1}}{a_n} < L + \epsilon$$

2. **Establishing the Ratio $q$:** Let we define $q = L + \epsilon$. Since we are given that $L < 1$, we can choose a sufficiently small $\epsilon$ such that $q < 1$. From this, we obtain the inequality:

$$a_{n+1} < a_n(L + \epsilon) = a_n q \quad \text{for all } n \geq N$$

3. **Inductive Expansion:** By applying the above inequality repeatedly for terms following $a_N$, we find:
   - For $n = N$: $a_{N+1} < a_N \cdot q$
   - For $n = N + 1$: $a_{N+2} < a_{N+1} \cdot q < (a_N \cdot q) \cdot q = a_N \cdot q^2$
   - In general, for any $k \geq 1$: $a_{N+k} < a_N \cdot q^k$

4. **Comparison to a Geometric Series:** The "tail" of the series starting from $N$ can be bounded as follows:

$$\sum_{k=0}^{\infty} a_{N+k} < \sum_{k=0}^{\infty} a_N \cdot q^k = a_N \sum_{k=0}^{\infty} q^k$$

5. **Conclusion:** The expression $\sum_{k=0}^{\infty} q^k$ is a geometric series. Since $|q| < 1$, this series converges. By the **Direct Comparison Test**, because the tail of our series is bounded by a convergent geometric series, the tail $\sum_{n=N}^{\infty} a_n$ converges. Since the sum of a finite part and a convergent tail is convergent, the original series $\sum_{n=1}^{\infty} a_n$ converges.

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## Exercise 2.7

**Question:** Does the condition $a_n > 0$ and $\overline{\lim}_{n\to\infty} \frac{a_{n+1}}{a_n} = q > 1$ imply that the series $\sum_{n=1}^{\infty} a_n$ diverges?

## Solution

To determine this, we examine the counter-example suggested in the hint:

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{2^3} + \frac{1}{3^3} + \dots$$

### 1. Define the Series

We define the terms of the series as follows:

- For odd terms ($n = 2k - 1$): $a_{2k-1} = \frac{1}{2^k}$
- For even terms ($n = 2k$): $a_{2k} = \frac{1}{3^k}$

### 2. Analyze the Ratios

We investigate the ratio $\frac{a_{n+1}}{a_n}$ by examining two cases:

1. **Case 1: $n$ is even ($n = 2k$)** In this case, the next term is odd ($a_{2k+1} = \frac{1}{2^{k+1}}$):

$$\frac{a_{2k+1}}{a_{2k}} = \frac{1/2^{k+1}}{1/3^k} = \frac{3^k}{2 \cdot 2^k} = \frac{1}{2}\left(\frac{3}{2}\right)^k$$

As $k \to \infty$, this ratio approaches $\infty$:

$$\lim_{k\to\infty} \frac{1}{2}\left(\frac{3}{2}\right)^k = \infty$$

2. **Case 2: $n$ is odd ($n = 2k - 1$)** In this case, the next term is even ($a_{2k} = \frac{1}{3^k}$):

$$\frac{a_{2k}}{a_{2k-1}} = \frac{1/3^k}{1/2^k} = \left(\frac{2}{3}\right)^k$$

As $k \to \infty$, this ratio approaches 0:

$$\lim_{k\to\infty} \left(\frac{2}{3}\right)^k = 0$$

### 3. Evaluate the Limit Superior

By the definition of the limit superior ($\overline{\lim}$), we look for the supremum of the limits of all convergent subsequences. Since one subsequence of ratios goes to $\infty$:

$$q = \overline{\lim}_{n\to\infty} \frac{a_{n+1}}{a_n} = \infty$$

Clearly, $q = \infty > 1$, which satisfies the condition of the question.

### 4. Check for Convergence

However, the series itself is the sum of two convergent geometric series:

$$\sum_{n=1}^{\infty} a_n = \sum_{k=1}^{\infty} \frac{1}{2^k} + \sum_{k=1}^{\infty} \frac{1}{3^k}$$

Using the sum formula $S = \frac{a_1}{1-r}$:

- $\sum_{k=1}^{\infty} \frac{1}{2^k} = \frac{1/2}{1-1/2} = 1$
- $\sum_{k=1}^{\infty} \frac{1}{3^k} = \frac{1/3}{1-1/3} = \frac{1}{2}$

The total sum is $1 + 0.5 = 1.5$. Since the sum is finite, the series **converges**.

### Conclusion

No, the condition $q > 1$ for the limit superior does not imply divergence. This demonstrates that while a limit $q > 1$ in D'Alembert's Test implies divergence, the limit superior version only guarantees convergence if $q < 1$.

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## Exercise 2.8: The Root Test

**Theorem:** Let $\sum_{n=1}^{\infty} a_n$ be a series with $a_n > 0$. Let $q = \overline{\lim}_{n\to\infty} \sqrt[n]{a_n}$.

1. If $q < 1$, the series $\sum_{n=1}^{\infty} a_n$ converges.
2. If $q > 1$, the series $\sum_{n=1}^{\infty} a_n$ diverges.

### Proof

**Case 1: $q < 1$ (Convergence)**

The condition $q < 1$ implies that for almost all terms of the sequence:

$$\sqrt[n]{a_n} < q + \epsilon$$

By choosing a sufficiently small $\epsilon$, we can define $r = q+\epsilon$ such that $r < 1$. Consequently, for almost all $n$ (specifically for all $n > N$):

$$a_n < r^n$$

Since $\sum r^n$ is a convergent geometric series (because $r < 1$), the series $\sum a_n$ converges by the Comparison Test.

**Case 2: $q > 1$ (Divergence)**

The condition $q > 1$ implies that there are an infinite number of terms for which:

$$\sqrt[n]{a_n} > 1$$

Raising both sides to the power of $n$ gives $a_n > 1$ for infinitely many values of $n$. Because there is an infinite subsequence of terms that are greater than 1, the general term $a_n$ does not approach 0 as $n \to \infty$. Since the necessary condition for convergence ($\lim_{n\to\infty} a_n = 0$) is not satisfied, the series diverges.

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## Exercise 2.9: Investigation of Convergence

**Problem:** Analyze the convergence of the series $\sum_{n=1}^{\infty} a_n$, where:

$$a_n = \frac{n^3(\sqrt{2} + (-1)^n)^n}{3^n}$$

### Solution

To investigate this series, we utilize the **Root Test** (Cauchy Test) because the general term is raised to the power of $n$. We calculate the limit superior:

$$q = \overline{\lim}_{n\to\infty} \sqrt[n]{a_n}$$

1. **Calculate the $n$-th root:**

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{n^3(\sqrt{2} + (-1)^n)^n}{3^n}} = \frac{(\sqrt[n]{n})^3 \cdot (\sqrt{2} + (-1)^n)}{3}$$

2. **Analyze the components as $n \to \infty$:**
   - We know that $\lim_{n\to\infty} \sqrt[n]{n} = 1$, which implies $(\sqrt[n]{n})^3 \to 1$.
   - The term $(\sqrt{2} + (-1)^n)$ oscillates based on whether $n$ is even or odd:
     - If $n$ is even: $\sqrt{2} + 1 \approx 2.414$

  - If $n$ is odd: $\sqrt{2}-1 \approx 0.414$

3. **Determine the Limit Superior ($q$):** The limit superior is the largest limit point of the sequence. This occurs when $n$ is even:
$$q = \overline{\lim}_{n\to\infty} \frac{(\sqrt[n]{n})^3 \cdot (\sqrt{2}+(-1)^n)}{3} = \frac{1\cdot(\sqrt{2}+1)}{3}$$
$$q = \frac{\sqrt{2}+1}{3} \approx \frac{2.414}{3} \approx 0.804$$

4. **Conclusion:** Since $q \approx 0.804 < 1$, the series satisfies the condition for convergence under the Root Test. Therefore, the series $\sum_{n=1}^{\infty} \frac{n^3(\sqrt{2}+(-1)^n)^n}{3^n}$ **converges**.

# Exercise 2.10: Riemann Rearrangement Theorem

**Problem:** Given the alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$, which converges conditionally, describe a way to rearrange its terms and list the first 11 terms of the new arrangement such that:

(a) The series converges to $\frac{\pi}{2} \approx 1.5708$.

(b) The series does not converge, even in the broad sense (it oscillates).

## Part (a): Converging to $\frac{\pi}{2}$

**Method:** Select positive terms in order until the partial sum first exceeds $\frac{\pi}{2}$. Then, add negative terms in order until the partial sum falls below $\frac{\pi}{2}$. Repeat this process indefinitely.

**First 11 terms:** The first partial sum exceeding $\frac{\pi}{2} \approx 1.5708$ is $S_4$:
$$1+\frac{1}{3}+\frac{1}{5}+\frac{1}{7} \approx 1.676$$

To move back toward the target, we add the first negative term:
$$1+\frac{1}{3}+\frac{1}{5}+\frac{1}{7}-\frac{1}{2} \approx 1.176$$

Now we add positive terms until we exceed $\frac{\pi}{2}$ again:
$$1+\frac{1}{3}+\frac{1}{5}+\frac{1}{7}-\frac{1}{2}+\frac{1}{9}+\frac{1}{11}+\frac{1}{13}+\frac{1}{15}+\frac{1}{17} \approx 1.581$$

Adding the next negative term:
$$1+\frac{1}{3}+\frac{1}{5}+\frac{1}{7}-\frac{1}{2}+\frac{1}{9}+\frac{1}{11}+\frac{1}{13}+\frac{1}{15}+\frac{1}{17}-\frac{1}{4} \approx 1.331$$

The first 11 terms are:
$$1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, -\frac{1}{2}, \frac{1}{9}, \frac{1}{11}, \frac{1}{13}, \frac{1}{15}, \frac{1}{17}, -\frac{1}{4}$$

## Part (b): Divergence through Oscillation

**Method:** Arrange terms such that the partial sums oscillate between values (e.g., above 1 and below 0) without settling.

**First 11 terms:** 1. Start with positive terms to exceed 1: $1+\frac{1}{3} \approx 1.333$. 2. Add the first negative term to drop: $1+\frac{1}{3}-\frac{1}{2} \approx 0.833$. 3. Add the next positive term to exceed 1: $1+\frac{1}{3}-\frac{1}{2}+\frac{1}{5} \approx 1.033$. 4. Add enough negative terms to drop below 0:
$$1, \frac{1}{3}, -\frac{1}{2}, \frac{1}{5}, -\frac{1}{4}, -\frac{1}{6}, -\frac{1}{8}, -\frac{1}{10}, -\frac{1}{12}, -\frac{1}{14}, -\frac{1}{16} \cdots \approx -0.17$$

The first 11 terms are:
$$1, \frac{1}{3}, -\frac{1}{2}, \frac{1}{5}, -\frac{1}{4}, -\frac{1}{6}, -\frac{1}{8}, -\frac{1}{10}, -\frac{1}{12}, -\frac{1}{14}, -\frac{1}{16}$$

## Exercise 2.11: Derivation of the Error Inequality

Given the alternating series:

$$S = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n^2+1}}$$

### 1. Identifying the Terms

The series is of the form $\sum(-1)^{n+1}b_n$, where the positive part of the sequence is:

$$b_n = \frac{1}{\sqrt{n^2+1}}$$

### 2. Applying the Alternating Series Estimation Theorem

The error (remainder) after summing $N$ terms is bounded by the first term we do *not* include in the sum:

$$|R_N| = |S - S_N| \leq b_{N+1}$$

To find $b_{N+1}$, we substitute $n = N+1$ into our expression for $b_n$:

$$b_{N+1} = \frac{1}{\sqrt{(N+1)^2+1}}$$

### 3. Setting the Accuracy Constraint

We are given that the sum must be accurate to within $10^{-5}$. This means we require the error to be strictly less than this threshold:

$$|R_N| < 10^{-5}$$

By transitivity, we obtain the governing inequality:

$$\frac{1}{\sqrt{(N+1)^2+1}} < 10^{-5}$$

### 4. Solving for N

Inverting both sides:

$$\sqrt{(N+1)^2+1} > 10^5$$

Squaring both sides:

$$(N+1)^2 + 1 > 10^{10}$$

$$(N+1)^2 > 10^{10} - 1$$

Taking the square root and solving for $N$:

$$N + 1 > \sqrt{9,999,999,999} \approx 100,000$$

$$N > 99,999 \implies N \geq 100,000$$